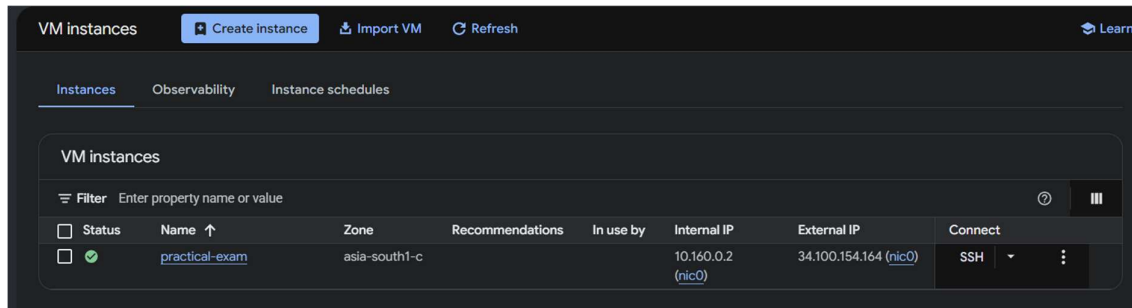


Cloud Computing Practical Examination

Part A – Cloud & Linux VM Setup

Task 1: Create a Linux Virtual Machine with Public IP and SSH access enabled.



Part B – Linux Commands

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ pwd
/home/nandini-sarukte
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ hostname
nandini-sarukte-VMware-Virtual-Platform
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ mkdir script
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ cd script
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch file1.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch file2.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch file3.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch filr4.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch file5.txt
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ mv filr4.txt file4.txt
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ rm file5.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ ls -a
.  ..  file1.txt  file2.txt  file3.txt  file4.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ ps
  PID TTY          TIME CMD
  4053 pts/0    00:00:00 bash
  4352 pts/0    00:00:00 ps
```

```

nandini-sarukte@nandini-sarukte-Vmware-Virtual-Platform:~/script$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.5  0.3 25592 16808 ?        Ss   17:19   0:07 /usr/lib/systemd/systemd --switched-root --system --deserialize=51 auto automati
root         2  0.0  0.0      0     0 ?        S    17:19   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    17:19   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-rcu_gp]
root         5  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-sync_wq]
root         6  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-netns]
root         9  0.1  0.0      0     0 ?        I    17:19   0:02 [kworker/0:0-events]
root        10  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/0:0H-kblockd]
root        12  0.0  0.0      0     0 ?        I    17:19   0:00 [kworker/u512:0-ipv6_addrconf]
root        13  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-nm_percpu_wq]
root        14  0.0  0.0      0     0 ?        S    17:19   0:00 [ksoftirqd/0]
root        15  0.0  0.0      0     0 ?        I    17:19   0:00 [rcu_preempt]
root        16  0.0  0.0      0     0 ?        S    17:19   0:00 [rcu_exp_par_gp_kthread_worker/1]
root        17  0.0  0.0      0     0 ?        S    17:19   0:00 [rcu_exp_gp_kthread_worker]
root        18  0.0  0.0      0     0 ?        S    17:19   0:00 [migration/0]
root        19  0.0  0.0      0     0 ?        S    17:19   0:00 [kprobe-optimizer]
root        20  0.0  0.0      0     0 ?        S    17:19   0:00 [idle_inject/0]
root        21  0.0  0.0      0     0 ?        S    17:19   0:00 [cpuhp/0]
root        22  0.0  0.0      0     0 ?        S    17:19   0:00 [cpuhp/1]
root        23  0.0  0.0      0     0 ?        S    17:19   0:00 [idle_inject/1]
root        24  0.0  0.0      0     0 ?        S    17:19   0:00 [migration/1]
root        25  0.0  0.0      0     0 ?        S    17:19   0:00 [ksoftirqd/1]
root        28  0.0  0.0      0     0 ?        I    17:19   0:00 [kworker/u513:0-events_power_efficient]
root        30  0.0  0.0      0     0 ?        S    17:19   0:00 [kdevtmpfs]
root        31  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-inet_frag_wq]
root        32  0.0  0.0      0     0 ?        I    17:19   0:00 [rcu_tasks_kthread]
root        33  0.0  0.0      0     0 ?        I    17:19   0:00 [rcu_tasks_rude_kthread]
root        34  0.0  0.0      0     0 ?        S    17:19   0:00 [kauditd]
root        35  0.0  0.0      0     0 ?        S    17:19   0:00 [khungtaskd]
root        36  0.0  0.0      0     0 ?        S    17:19   0:00 [oom_reaper]
root        38  0.0  0.0      0     0 ?        I    17:19   0:00 [kworker/u513:2-events_unbound]
root        39  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-writeback]
root        40  0.0  0.0      0     0 ?        S    17:19   0:00 [kcompactd0]
root        41  0.0  0.0      0     0 ?        SN   17:19   0:00 [ksmd]
root        42  0.0  0.0      0     0 ?        SN   17:19   0:00 [khugepaged]
root        43  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-kblockd]
root        44  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-blkcg_punt_bio]
root        45  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-kintegrityd]
root        46  0.0  0.0      0     0 ?        S    17:19   0:00 [irq/9-acpi]
root        47  0.1  0.0      0     0 ?        I    17:19   0:01 [kworker/1:1-events]
root        48  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-tpm_dev_wq]
root        49  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-ata_sff]
root        50  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-md_bitmap]
root        51  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-md_l1bitmap_io]
root        52  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-md_l1bitmap_unplug]
root        53  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-md]
root        54  0.0  0.0      0     0 ?        I<   17:19   0:00 [kworker/R-edac-poller]

```

```

top - 17:30:43 up 10 min, 1 user, load average: 0.23, 0.33, 0.35
Tasks: 293 total, 1 running, 292 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.9 us, 6.2 sy, 0.0 ni, 85.8 id, 0.0 wa, 0.0 hi, 0.2 si, 0.0 st
MiB Mem : 5360.7 total, 2909.1 free, 1270.0 used, 1437.1 buff/cache
MiB Swap: 3514.0 total, 3514.0 free, 0.0 used, 4090.7 avail Mem

```

PID	USER	PR	NI	VRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3005	nandini+	20	0	3752232	239116	139564	S	17.9	4.4	0:29.81	gnome-shell
4019	nandini+	20	0	1612140	233984	117240	S	13.3	4.3	0:15.06	ptyxis
38	root	20	0	0	0	0	I	0.7	0.0	0:00.29	kworker/u513:2-events_unbound
1149	root	-51	0	0	0	0	S	0.7	0.0	0:00.64	irq/16-vmmgfx
47	root	20	0	0	0	0	I	0.3	0.0	0:01.02	kworker/1:1-events
453	root	20	0	0	0	0	I	0.3	0.0	0:00.48	kworker/u514:26-events_power_efficient
4344	nandini+	20	0	22264	6520	4252	R	0.3	0.1	0:00.08	top
1	root	20	0	25592	16808	11736	S	0.0	0.3	0:07.60	systend
2	root	20	0	0	0	0	S	0.0	0.0	0:00.06	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slab_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.82	kworker/0:0-events
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u512:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.04	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.54	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.01	rcu_exp_par_gp_kthread_worker/1
17	root	20	0	0	0	0	S	0.0	0.0	0:00.02	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.02	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.03	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:00.05	ksoftirqd/1
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-kblockd
28	root	20	0	0	0	0	I	0.0	0.0	0:00.66	kworker/u513:0-flush-8:0
30	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kdevtmpfs
31	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_frag_wq
32	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
33	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
34	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
35	root	20	0	0	0	0	S	0.0	0.0	0:00.00	khungtaskd
36	root	20	0	0	0	0	S	0.0	0.0	0:00.00	oom_reaper
39	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-writeback
40	root	20	0	0	0	0	S	0.0	0.0	0:00.12	kcompactd0
41	root	25	5	0	0	0	S	0.0	0.0	0:00.00	ksmd
42	root	39	19	0	0	0	S	0.0	0.0	0:00.00	khugepaged
43	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kblockd
44	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-blkcg_punt_bio
45	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kintegrityd

```

root      60 0.0 0.0 0 0 ? S 17:19 0:00 [ecryptfs-kthread]
root      61 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-kthrotld]
root      62 0.0 0.0 0 0 ? S 17:19 0:00 [irq/24-pciehp]
root      63 0.0 0.0 0 0 ? S 17:19 0:00 [irq/25-pciehp]
root      64 0.0 0.0 0 0 ? S 17:19 0:00 [irq/26-pciehp]
root      65 0.0 0.0 0 0 ? S 17:19 0:00 [irq/27-pciehp]
root      66 0.0 0.0 0 0 ? S 17:19 0:00 [irq/28-pciehp]
root      67 0.0 0.0 0 0 ? S 17:19 0:00 [irq/29-pciehp]
root      68 0.0 0.0 0 0 ? S 17:19 0:00 [irq/30-pciehp]
root      69 0.0 0.0 0 0 ? S 17:19 0:00 [irq/31-pciehp]
root      70 0.0 0.0 0 0 ? S 17:19 0:00 [irq/32-pciehp]
root      71 0.0 0.0 0 0 ? S 17:19 0:00 [irq/33-pciehp]
root      72 0.0 0.0 0 0 ? S 17:19 0:00 [irq/34-pciehp]
root      73 0.0 0.0 0 0 ? S 17:19 0:00 [irq/35-pciehp]
root      74 0.0 0.0 0 0 ? S 17:19 0:00 [irq/36-pciehp]
root      75 0.0 0.0 0 0 ? S 17:19 0:00 [irq/37-pciehp]
root      76 0.0 0.0 0 0 ? S 17:19 0:00 [irq/38-pciehp]
root      77 0.0 0.0 0 0 ? S 17:19 0:00 [irq/39-pciehp]
root      78 0.0 0.0 0 0 ? S 17:19 0:00 [irq/40-pciehp]
root      79 0.0 0.0 0 0 ? S 17:19 0:00 [irq/41-pciehp]
root      80 0.0 0.0 0 0 ? S 17:19 0:00 [irq/42-pciehp]
root      81 0.0 0.0 0 0 ? S 17:19 0:00 [irq/43-pciehp]
root      82 0.0 0.0 0 0 ? S 17:19 0:00 [irq/44-pciehp]
root      83 0.0 0.0 0 0 ? S 17:19 0:00 [irq/45-pciehp]
root      84 0.0 0.0 0 0 ? S 17:19 0:00 [irq/46-pciehp]
root      85 0.0 0.0 0 0 ? S 17:19 0:00 [irq/47-pciehp]
root      86 0.0 0.0 0 0 ? S 17:19 0:00 [irq/48-pciehp]
root      87 0.0 0.0 0 0 ? S 17:19 0:00 [irq/49-pciehp]
root      88 0.0 0.0 0 0 ? S 17:19 0:00 [irq/50-pciehp]
root      89 0.0 0.0 0 0 ? S 17:19 0:00 [irq/51-pciehp]
root      90 0.0 0.0 0 0 ? S 17:19 0:00 [irq/52-pciehp]
Floppy Disk 91 0.0 0.0 0 0 ? S 17:19 0:00 [irq/53-pciehp]
root      92 0.0 0.0 0 0 ? S 17:19 0:00 [irq/54-pciehp]
root      93 0.0 0.0 0 0 ? S 17:19 0:00 [irq/55-pciehp]
root      94 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-acpi_thermal_pm]
root      96 0.0 0.0 0 0 ? S 17:19 0:00 [scsi_eh_0]
root      97 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-scsi_tmf_0]
root      98 0.0 0.0 0 0 ? S 17:19 0:00 [scsi_eh_1]
root      99 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-scsi_tmf_1]
root     103 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-mld]
root     104 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-ipv6_addrconf]
root     105 0.0 0.0 0 0 ? I 17:19 0:00 [kworker/u512:1-ipv6_addrconf]
root     106 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-kstrp]
root     109 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/u515:0]
root     111 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/u517:0-ttn]
root     122 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-charger_manager]
root     366 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-npt_poll_0]
root     367 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-npt_0]
root     368 0.0 0.0 0 0 ? S 17:19 0:00 [scsi_eh_2]
root     369 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-scsi_tmf_2]
root     370 0.0 0.0 0 0 ? S 17:19 0:00 [scsi_eh_3]
root     371 0.0 0.0 0 0 ? I< 17:19 0:00 [kworker/R-scsi_tmf_3]

```

Part C – Permissions & User

```

nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ touch secure_data.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ ch mod 640 secure_data.txt
ch: command not found
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ chmod 640 secure_data.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$

```

```

nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ ls -l secure_data.txt
-rw-r----- 1 nandini-sarukte nandini-sarukte 0 Jun 10 17:47 secure_data.txt
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$

```

```

nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$ id
uid=1000(nandini-sarukte) gid=1000(nandini-sarukte) groups=1000(nandini-sarukte),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users),111(lpadmin),114(lxd)
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~/script$

```


Part D – Shell Scripting

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ nano secure.sh
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ chmod +x secure.sh
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ ./secure.sh
Welcome to cloud computing
Show Apps nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$
```

```
#!/bin/bash
echo "Enter your username"
read username
echo "Username is $username"
```

```
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ chmod +x secure.sh
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$ ./secure.sh
Enter your username
nandini-sarukte
Username is nandini-sarukte
nandini-sarukte@nandini-sarukte-VMware-Virtual-Platform:~$
```